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NAAC GRADE A+ ACCREDITED UNIVERSITY

Worksheet 8

Student Name:Aryan Tomar **UID:25MCI10167**
Branch:MCA (AI&ML) **Section/Group:1/A**
Semester:2nd **Date of Performance:02/04/2026**
Subject Name:- DBMS LAB **Subject Code: 25CAP-652**

1. Aim of the Session

To apply the concept of Stored Procedures in database operations in order to perform tasks like insertion,updating, deletion, and retrieval of data efficiently, securely, and in a reusable manner within the database system.

2. Software Requirements

- PostgreSQL (Database Server)
- pgAdmin
- Windows Operating System

3. Objective of the Session

To understand and implement stored procedures for performing database operations like insertion, updating, deletion, and retrieval in an efficient and reusable manner.

4. Practical / Experiment Steps

Prerequisite Understanding

• Students should understand basic SQL operations such as SELECT, INSERT, UPDATE, and DELETE.

• Students should be familiar with table creation and data types in PostgreSQL.

• Students should understand the concept of functions and procedures in programming.

• Students should have basic knowledge of PL/pgSQL blocks (BEGIN, END, DECLARE).

• Students should understand parameters (IN, OUT, INOUT) used in stored procedures.

• Students should be aware of basic conditional statements like IF–ELSE and exception handling.

5. Procedure of the Practical

(i) Start the system and log in to the computer.

(ii) Open PostgreSQL software.

iii) Create and select the database.

Create database Practical8;

(iv) Create table using DDL command.

```
CREATE TABLE Employees2 (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    manager_id INT,
    department VARCHAR(50),
    salary INT
);
```

(v) Insert records into the table.

INSERT INTO Employees2 VALUES

(1, 'Amit', NULL, 'Management', 120000),

(2, 'Ravi', 1, 'Engineering', 80000),

(3, 'Neha', 1, 'Engineering', 82000),

(4, 'Karan', 2, 'Engineering', 60000),

(5, 'Simran', 2, 'Engineering', 62000),

(6, 'Pooja', 3, 'Engineering', 61000),

(7, 'Rahul', 3, 'Engineering', 64000),

(8, 'Arjun', 1, 'HR', 70000);

(vi) Display all records.

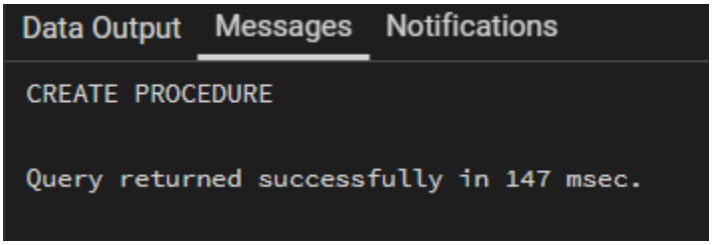
select * from employee;

	emp_id [PK] integer	emp_name character varying (50)	manager_id integer	department character varying (50)	salary integer
1	1	Amit	[null]	Management	120000
2	2	Ravi	1	Engineering	80000
3	3	Neha	1	Engineering	82000
4	4	Karan	2	Engineering	60000
5	5	Simran	2	Engineering	62000
6	6	Pooja	3	Engineering	61000
7	7	Rahul	3	Engineering	64000
8	8	Arjun	1	HR	70000

Step 1: Creating a procedure

```
create or replace procedure update_salary_procc(IN p_emp_id int,inout p_salary
numeric(20,3),OUT status varchar(20))

AS

$$

declare

curr_sal numeric(20,3);

begin

select salary+p_salary into curr_sal from Employees2 where emp_id=p_emp_id;

if not found then

raise exception 'employee not found';

end if;



update Employees2

set salary=curr_sal

where emp_id=p_emp_id;

p_salary:=curr_sal;

status:='success';

exception

when others then

if sqlerrm like '%employee not found%' then
```

```
status:='employee not found';
end if;
end;
$$ language plpgsql;
```



Step 2: Execute Procedure (Invalid Employee Case)

```
do
$$
declare
emp_id int :=99;
status varchar(20);
salary numeric(20,3):=500;
begin
call update_salary_procc(emp_id,salary,status);
raise notice 'your status is %',status;
end ;
$$
```

```
Data Output   Messages   Notifications

NOTICE:  your status is employee not found
DO

Query returned successfully in 87 msec.
```

Step 3: Execute Procedure (Valid Employee Case)

```
do
$$
declare
emp_id int := 1;
status varchar(20);
salary numeric(20,3) := 500;
begin
call update_salary_procc(emp_id, salary, status);
raise notice 'your status is %', status;
end;
$$;
```

```
NOTICE:  your status is success
DO

Query returned successfully in 76 msec.
```

Step4 :Verify Updated Data

select * from Employees2;

	emp_id [PK] integer	emp_name character varying (50)	manager_id integer	department character varying (50)	salary integer
1	2	Ravi	1	Engineering	80000
2	3	Neha	1	Engineering	82000
3	4	Karan	2	Engineering	60000
4	5	Simran	2	Engineering	62000
5	6	Pooja	3	Engineering	61000
6	7	Rahul	3	Engineering	64000
7	8	Arjun	1	HR	70000
8	1	Amit	[null]	Management	120500

6. I/O Analysis (Input / Output)

Input:

• Employee records inserted into the Employees2 table

• Stored procedure definition with IN, OUT, and INOUT parameters

• Employee ID and salary increment value passed as input

• PL/pgSQL DO blocks used to call the procedure

• Conditional logic and exception handling inside the procedure

Output:

• Salary updated successfully for valid employee IDs
• Error message handled when employee is not found
• Status message returned as **'success'** or **'employee not found'**
• Updated salary reflected in the Employees2 table

• Proper execution of stored procedure with parameter passing
• Reusable business logic implemented inside the database

7. Learning Outcomes

• Understood the concept and use of stored procedures in PostgreSQL

• Learned how to use IN, OUT, and INOUT parameters effectively

• Gained experience in writing reusable and efficient database logic

• Understood exception handling using PL/pgSQL

• Learned how to perform conditional updates within procedures

• Developed skills to implement business logic at the database level

• Improved understanding of procedural SQL for real-world applications